Cardiovascular System (CARD) · NPLEX-P1-CARD-18

CV-18-A Hypertensive Heart Diseases: Systemic and Pulmonary Hypertension

Work through each activity on blank paper with the notes chapter closed, then open the answer key and mark what you missed. Do the two brain dumps for a competency on different days rather than back to back.

Work each activity without your notes first. Rate your confidence before you check. Then read the notes, write down what you missed, and set a date to come back to it.

Competency tracker

Competency	BD1 done	BD2 done	Confidence 1 to 4	Next review date
NPLEX-P1-CARD-18-01-C01				
NPLEX-P1-CARD-18-01-C02				
NPLEX-P1-CARD-18-01-C03				
NPLEX-P1-CARD-18-01-C04				
NPLEX-P1-CARD-18-01-C05				
NPLEX-P1-CARD-18-01-C06				

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-01-C01-BD1

Blood pressure categories and why the kidney sets pressure

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-A Hypertensive Heart Diseases: Systemic and Pulmonary Hypertension Competency: NPLEX-P1-CARD-18-01-C01 Activity: NPLEX-P1-CARD-18-01-C01-BD1 Date: _____

Without looking at your notes, write out the four blood pressure categories of the 2017 ACC/AHA scheme with the systolic and diastolic values for each, and write the rule that applies when the two values fall in different categories. Beside that, write how many readings and occasions the diagnosis needs. Then write the equation that relates mean arterial pressure to cardiac output and systemic vascular resistance, and state which of the two is raised in established primary hypertension. Underneath, explain in four or five sentences why baroreceptors cannot hold pressure high over months and how pressure natriuresis works, and finish by writing the one change in the kidney that every sustained hypertension must involve.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-01-C01-BD2

Office readings, night-time pressure, and two inherited salt channels

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-A Hypertensive Heart Diseases: Systemic and Pulmonary Hypertension Competency: NPLEX-P1-CARD-18-01-C01 Activity: NPLEX-P1-CARD-18-01-C01-BD2 Date: _____

Three patients are measured. The first reads 128/86 mmHg in clinic and 118/74 mmHg at home, the second reads 118/76 mmHg in clinic and 142/90 mmHg at home, and the third reads 150/94 mmHg in clinic with no fall in pressure during sleep. For each one, name what the pattern is called, state whether the ACC/AHA category assigned in clinic is the right one, and write what the pattern suggests about the patient. Then take a 19-year-old with hypertension, a plasma potassium of 2.9 mEq/L (reference range 3.5 to 5.0 mEq/L), metabolic alkalosis, and low renin with low aldosterone. Write the two inherited disorders and the one dietary exposure from the notes that produce this picture, give the protein or enzyme altered in each, and explain step by step why each one raises sodium reabsorption, wastes potassium, and leaves both renin and aldosterone low. Finish by writing which risk factors in the notes act partly by raising sympathetic outflow and which act mainly by expanding volume.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-01-C02-BD1

Build the secondary hypertension table from memory

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-A Hypertensive Heart Diseases: Systemic and Pulmonary Hypertension Competency: NPLEX-P1-CARD-18-01-C02 Activity: NPLEX-P1-CARD-18-01-C02-BD1 Date: _____

On blank paper build a table of the secondary causes of hypertension with one row each for renal artery stenosis, primary aldosteronism, the mineralocorticoid group of Cushing syndrome with licorice and Liddle syndrome, pheochromocytoma, coarctation of the aorta, and obstructive sleep apnea. Use the columns key clinical clue, renin, and aldosterone, and fill every cell. Beside the renal artery stenosis row, write the two diseases that narrow the artery, the age and sex each one affects, and the part of the artery each one involves. Beside the pheochromocytoma row, write which receptors raise resistance and which raise cardiac output. Then write the share of all hypertension that is secondary and list the four features of a patient that should make you look for a cause.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-01-C02-BD2

Reason from the laboratory pattern back to the organ

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-A Hypertensive Heart Diseases: Systemic and Pulmonary Hypertension Competency: NPLEX-P1-CARD-18-01-C02 Activity: NPLEX-P1-CARD-18-01-C02-BD2 Date: _____

Four patients are hypertensive and hypokalemic. Patient A has high renin and high aldosterone with an abdominal bruit. Patient B has low renin and high aldosterone with no edema. Patient C has low renin and low aldosterone and drinks licorice tea daily. Patient D has normal renin with episodic headache, sweating, and palpitations. For each patient write the organ or vessel at fault and explain in two or three sentences why renin is where it is and why aldosterone is where it is. For patient B, explain the tubular steps that produce the low potassium and the alkalosis, and explain why the patient has no edema despite retaining sodium. For patient C, name the enzyme involved and write which receptor is being occupied and by which hormone. For patient D, write why the plasma metanephrines stay high between attacks and why an epinephrine-secreting tumor is almost always inside the adrenal gland. Finish by writing what happens to the glomerular filtration rate of a stenotic kidney when angiotensin II is blocked, and why.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Comparison table NPLEX-P1-CARD-18-01-C02-CP01

Renal artery stenosis against primary aldosteronism

Fill in the table from memory, then check it against the notes and correct any cell you missed.

Feature	Renal artery stenosis	Primary aldosteronism
Where the primary abnormality sits		
Plasma renin and why it is at that level		
Plasma aldosterone and why it is at that level		
Plasma potassium and acid-base state		
Physical examination and imaging clues		
Whether edema develops, and the reason		
What happens to the kidney when angiotensin II is blocked		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-01-C03-BD1

Hyaline arterioles and the concentrically hypertrophied ventricle

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-A Hypertensive Heart Diseases: Systemic and Pulmonary Hypertension Competency: NPLEX-P1-CARD-18-01-C03 Activity: NPLEX-P1-CARD-18-01-C03-BD1 Date: _____

Write the law of Laplace as the notes give it, with each term named, then explain in four or five sentences why a raised afterload makes myocytes add sarcomeres in parallel and what that does to wall stress. Write the thickness of a normal left ventricular wall and the thickness reached in hypertensive heart disease, and write how the sarcomere arrangement and the chamber size differ between pressure overload and volume overload. Underneath, describe hyaline arteriolosclerosis: what leaks into the wall, what the smooth muscle cells add, how the wall looks on routine stains, what happens to the lumen, and the two other conditions in which the same lesion appears. Finish by writing what happens to the myocyte nuclei in a hypertrophied wall.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-01-C03-BD2

Trace chronic pressure into four organs

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-A Hypertensive Heart Diseases: Systemic and Pulmonary Hypertension Competency: NPLEX-P1-CARD-18-01-C03 Activity: NPLEX-P1-CARD-18-01-C03-BD2 Date: _____

A 70-year-old woman has had untreated hypertension for 25 years. Her echocardiogram shows a left ventricular wall of 1.9 cm with an ejection fraction of 60%, and she has just developed atrial fibrillation with breathlessness at rest. Write why a ventricle with a normal ejection fraction can congest the lungs, and write what the loss of atrial contraction and the fast rate each take away from filling. Then write the extra heart sound this ventricle produces, which event in the cardiac cycle makes it, and the ECG findings in V1, V5, and V6 that reflect the increased mass, including the strain pattern. Next write why this thick wall is prone to ischemia even with open coronary arteries, naming three reasons from the notes. Then, for her kidneys, retinas, and brain, write what you expect to find: name the renal lesion with the appearance of the kidney surface and what happens to proteinuria and glomerular filtration rate, name five retinal findings and the mechanism of the one caused by a thick arteriole crossing a vein, and name the two lesions of penetrating cerebral arteries with the size and the usual sites of each. Finish by writing why hypertensive bleeds sit deep while amyloid angiopathy bleeds sit in the lobes.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-18-01-C03-RT01

Rebuild the pressure overload sequence

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-A Hypertensive Heart Diseases: Systemic and Pulmonary Hypertension Competency: NPLEX-P1-CARD-18-01-C03 Activity: NPLEX-P1-CARD-18-01-C03-RT01 Date: _____

Rebuild from memory the seven-step sequence that runs from sustained hypertension to atrial fibrillation and heart failure with a preserved ejection fraction, one step per line and in order. Beside the second step, write the law of Laplace and show which term rises first. Beside the third step, write where the new sarcomeres go and what happens to chamber volume. Beside the fifth step, write why a higher left atrial pressure becomes necessary. Then draw the left ventricle and left atrium in cross section at the start of the sequence and at the end, label the wall thickness and the atrial size on each drawing, and write one sentence on what can happen to the chamber and to systolic function very late in the course.

Rebuild from memory: Pressure overload to diastolic failure and atrial fibrillation

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-01-C04-BD1

Autoregulation, breakthrough, and the acute arteriolar lesion

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-A Hypertensive Heart Diseases: Systemic and Pulmonary Hypertension Competency: NPLEX-P1-CARD-18-01-C04 Activity: NPLEX-P1-CARD-18-01-C04-BD1 Date: _____

Write the definition of a hypertensive emergency with the pressure usually quoted, then write how it differs from hypertensive urgency and what the older term malignant hypertension described. Write which matters more, the level reached or the rate of rise. Draw the cerebral autoregulation curve with mean arterial pressure on the horizontal axis and cerebral blood flow on the vertical axis, mark the lower and upper limits in a healthy person, and draw a second curve showing what chronic hypertension does to it. On the drawing, mark the point of breakthrough and write in five or six sentences what happens to the arterioles, the capillaries, and the blood-brain barrier beyond that point, the symptoms this produces, and which part of the brain the edema favors and why. Underneath, describe the acute arteriolar lesion: what fibrinoid necrosis is and how it looks on routine stains, what the surviving arterioles build and what that lesion is called, where in the kidney it is most conspicuous, and one other disease that produces the same lesion.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-01-C04-BD2

Why the lower pressure did more damage

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-A Hypertensive Heart Diseases: Systemic and Pulmonary Hypertension Competency: NPLEX-P1-CARD-18-01-C04 Activity: NPLEX-P1-CARD-18-01-C04-BD2 Date: _____

A 72-year-old woman with 30 years of hypertension has a pressure of 186/112 mmHg, no symptoms, and a normal fundus. A previously normotensive 24-year-old woman at 36 weeks of pregnancy has a pressure of 168/108 mmHg with headache and confusion. Write which of the two has a hypertensive emergency and which has urgency, then explain from the position of each woman's autoregulation curve why the lower pressure caused acute brain injury. Next, the younger woman's creatinine rises, her urine shows blood, her platelet count falls, and her blood film shows schistocytes with a raised lactate dehydrogenase. Write the arteriolar lesion behind each of those findings and explain how fibrin in a damaged small vessel produces that blood film. Then write out the renal vicious cycle in this setting in six steps and explain why the renin level is very high here while it is variable in chronic hypertension, and why a shrinking plasma volume pushes renin higher still. Finish by listing the triggers from the notes that precipitate an emergency, and write why stopping clonidine abruptly belongs on that list.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Prediction exercise NPLEX-P1-CARD-18-01-C04-PR01

Predict what a rapid severe rise in pressure does

A previously normotensive 30-year-old develops acute glomerulonephritis and his pressure rises from 118/72 mmHg to 196/124 mmHg over three days. Predict each item below, write the direction of the change, and give the mechanism that explains it in one line.

What changes	Up, down, or no change	Why, in one or two lines
What happens to cerebral arteriolar constriction once mean arterial pressure passes the upper limit of autoregulation		
What happens to the permeability of the blood-brain barrier and to the water content of the brain		
What the renal arteriolar walls look like on biopsy within days		
What happens to renin release from the juxtaglomerular cells and why		
What happens to the platelet count, the blood film, and the lactate dehydrogenase		
What the kidney surface looks like at this stage and how that differs from the kidney of long-standing moderate hypertension		
What happens to the left ventricle acutely, given that it has had no time to hypertrophy		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-01-C05-BD1

Pulmonary pressures, the resistance equation, and the five groups

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-A Hypertensive Heart Diseases: Systemic and Pulmonary Hypertension Competency: NPLEX-P1-CARD-18-01-C05 Activity: NPLEX-P1-CARD-18-01-C05-BD1 Date: _____

Write the normal pulmonary artery pressure as systolic over diastolic with the mean value, and write how pulmonary vascular resistance compares with systemic resistance. Write what happens to pulmonary vascular resistance during exercise and the two mechanical reasons for it. Then write the catheter-based definition of pulmonary hypertension with the threshold the notes use and the older value many textbooks still quote, and write which test raises the suspicion first and which measurement it uses. Write the equation for pulmonary vascular resistance, name the units, and give the value of one of those units. Set out the three hemodynamic types with the wedge pressure and resistance values that define each. Finally, build the five-group classification from memory as a table with the main mechanism and two examples in each group, and mark which group is the most common cause and which group is the most common after it.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-01-C05-BD2

Separate two catheter studies and explain two mechanisms

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-A Hypertensive Heart Diseases: Systemic and Pulmonary Hypertension Competency: NPLEX-P1-CARD-18-01-C05 Activity: NPLEX-P1-CARD-18-01-C05-BD2 Date: _____

Two patients each have a mean pulmonary artery pressure of 36 mmHg and a cardiac output of 5 L/min. Patient A has a wedge pressure of 10 mmHg and patient B has a wedge pressure of 28 mmHg. Calculate the pulmonary vascular resistance for each, show your working, state which hemodynamic type each one has, and write two diseases that would produce each result. Write in two sentences why the same mean pressure points to different organs. Then take a patient with severe chronic obstructive pulmonary disease and write out, step by step, how low alveolar oxygen reaches pulmonary arterial smooth muscle contraction, naming the channels and the ion movements in order. Write why that response is useful when one region is poorly ventilated and harmful when the whole lung is hypoxic, and write the one blood change in chronic hypoxemia that adds to the pressure and how it does so. Finish by writing how long-standing mitral stenosis can add a precapillary component to a problem that began behind the left atrium, and how the perfusion scan of chronic thromboembolic disease differs from that of group 1.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-01-C06-BD1

Rebuild the pathogenesis of pulmonary arterial hypertension

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-A Hypertensive Heart Diseases: Systemic and Pulmonary Hypertension Competency: NPLEX-P1-CARD-18-01-C06 Activity: NPLEX-P1-CARD-18-01-C06-BD1 Date: _____

Name the gene most often mutated in heritable pulmonary arterial hypertension, the receptor family it belongs to, the signaling pathway it uses, and the two things that signaling normally does to smooth muscle. Write the inheritance pattern and why a second hit is thought to be needed. Then build a table of the three endothelial mediators from the notes with their receptor and second messenger, their effect on vessel diameter and on cell growth, and the direction each one moves in this disease. Underneath, rebuild the seven-step sequence from the initial endothelial injury to right ventricular failure, one step per line and in order. Beside the remodeling step, write the three structural changes in the small arteries in the order they appear and describe the lesion seen in severe disease, where it forms and what it resembles. Finish by listing the acquired triggers and associated diseases from the notes, and write what happens to a large left-to-right shunt once pulmonary resistance exceeds systemic resistance.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-01-C06-BD2

Explain the signs of the pressure-loaded right heart

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-A Hypertensive Heart Diseases: Systemic and Pulmonary Hypertension Competency: NPLEX-P1-CARD-18-01-C06 Activity: NPLEX-P1-CARD-18-01-C06-BD2 Date: _____

A patient with long-standing pulmonary arterial hypertension has a loud P2, a left parasternal heave, a right-sided fourth heart sound, a holosystolic murmur that gets louder on inspiration, large v waves in the jugular venous pulse, raised jugular venous pressure, hepatomegaly, and ankle edema. Her ECG shows right axis deviation, a tall R wave in V1, and peaked P waves. Take each finding in turn and write the mechanism that produces it. Then write the two stages the right ventricle passes through under a rising load and explain how dilation produces the murmur and what it does to the interventricular septum and to left ventricular filling. Write why the right ventricle becomes ischemic as its pressure approaches systemic pressure, naming what happens to its coronary perfusion. Write the definition of cor pulmonale, including what it excludes, and contrast the chronic form with the form that follows a massive pulmonary embolism. Finish by explaining why this patient's systemic blood pressure can be low although her left ventricle is structurally normal, and why syncope on exertion is an ominous finding.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Clinical application NPLEX-P1-CARD-18-01-C06-CA01

Progressive breathlessness in a young woman with a loud P2

A 32-year-old woman comes to the physician because of 10 months of worsening breathlessness on exertion and fatigue, and two episodes of fainting while climbing stairs. She has become hoarse over the past month. She takes no medicines, has never smoked, and has no history of liver disease, joint pain, or skin tightening. Her mother died at 38 years of age of an unexplained heart condition. Temperature is 36.8 °C (98.2 °F), pulse is 104/min, respirations are 22/min, and blood pressure is 96/62 mmHg. Jugular venous pressure is raised with large v waves, the second heart sound has a loud pulmonary component, there is a heave to the left of the sternum, and a holosystolic murmur at the lower left sternal border becomes louder on inspiration. The liver is enlarged and there is pitting edema of both ankles. The lungs are clear. The electrocardiogram shows right axis deviation, a tall R wave in V1, and peaked P waves in lead II. Right heart catheterization shows a mean pulmonary artery pressure of 52 mmHg, a wedge pressure of 9 mmHg, and a cardiac output of 3.4 L/min. Ventilation-perfusion scanning shows no mismatched perfusion defects.

1. Recognize. Name the hemodynamic type of pulmonary hypertension shown by the catheter numbers, calculate the pulmonary vascular resistance with your working, and name the classification group this patient belongs to.

2. Explain. Explain the molecular pathogenesis you expect in this patient, from the inherited defect her family history suggests through the endothelial mediator changes to the structural remodeling of the small arteries.

3. Differentiate. Write how the catheter numbers and the clinical picture separate this from pulmonary hypertension caused by left heart disease, by chronic lung disease, and by chronic thromboembolic obstruction, naming the finding that excludes each one.

4. Investigate. Write which further findings you would look for to support this group and to exclude the associated diseases listed in the notes, and what each finding would show.

5. Decide. State which chamber is failing, name the two physical findings on this examination that show it is failing rather than only hypertrophied, and name the term for right ventricular failure caused by pulmonary vascular disease.

6. Justify. Justify why her cardiac output is low and her systemic pressure is 96/62 mmHg although her left ventricle is structurally normal, why she fainted on exertion, and why she has become hoarse.

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question NPLEX-P1-CARD-18-01-C04-Q01

Arteriolar lesion in an abrupt severe rise in pressure

A 34-year-old man comes to the emergency department because of headache, blurred vision, and confusion that began 2 days ago. He was told his blood pressure was normal at a check 4 months ago. Blood pressure is 212/128 mmHg. Funduscopic examination shows flame-shaped hemorrhages, exudates, and papilledema. Serum creatinine is 3.2 mg/dL, urinalysis shows blood, the platelet count is 68,000/mm3, and the blood film shows fragmented erythrocytes. A kidney biopsy is performed. Which of the following is most likely to be seen in the renal arterioles?

A. Homogeneous glassy pink thickening of the wall with a narrowed lumen

B. Subendothelial lipid-laden macrophages with a fibrous cap

C. Concentric laminated layers of smooth muscle with fibrin deposition and wall necrosis

D. Nodular mesangial matrix expansion with capillary basement membrane thickening

Why did you pick it?

Board-style question NPLEX-P1-CARD-18-01-C05-Q02

Reading a right heart catheter study

A 68-year-old woman comes to the physician because of 6 months of breathlessness on exertion. Blood pressure is 148/88 mmHg. Right heart catheterization shows a mean pulmonary artery pressure of 34 mmHg, a pulmonary artery wedge pressure of 26 mmHg, and a cardiac output of 4 L/min. Echocardiography shows a left ventricular wall thickness of 1.8 cm, an ejection fraction of 60%, and an enlarged left atrium. Which of the following best accounts for her pulmonary artery pressure?

A. Backward transmission of a raised left atrial pressure

B. Proliferative remodeling of small pulmonary arteries with plexiform lesions

C. Hypoxic pulmonary vasoconstriction with loss of alveolar capillaries

D. Organized thrombus obstructing the proximal pulmonary arteries

Why did you pick it?